

A Proposal of Mach-Zehnder Interferometer

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I. INTRODUCTION

This proposal presents the design and simulation of an unbalanced Mach-Zehnder interferometer (MZI) for extracting the group index of a silicon waveguide. The waveguide and MZI will be simulated, different design parameters will be investigated, and the group index obtained from the MZI free spectral range (FSR) will be compared with the waveguide simulation.

II. THEORY

The group index of a waveguide is related to its wavelength-dependent effective index by

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \quad (1)$$

An unbalanced Mach-Zehnder interferometer (MZI) consists of two interferometer arms with a path-length difference $\Delta L = L_2 - L_1$. The phase difference between the two arms is

$$\Delta\phi = \frac{2\pi n_{\text{eff}} \Delta L}{\lambda} \quad (2)$$

The wavelength-dependent phase difference produces periodic maxima and minima in the MZI transmission spectrum. The wavelength spacing between two adjacent interference features is the free spectral range (FSR), which is related to the group index by

$$FSR \approx \frac{\lambda^2}{n_g \Delta L} \quad (3)$$

Thus, the group index can be extracted from the measured FSR using

$$n_g = \frac{\lambda^2}{FSR \Delta L} \quad (4)$$

III. MODELLING AND SIMULATION

A. Waveguide Modelling

In our design, a silicon strip waveguide with a width of 500 nm and a height of 220 nm is used. The waveguide is surrounded by SiO₂ cladding and is designed for quasi-TE polarization. The waveguide is modeled using Lumerical MODE with the finite-difference eigenmode (FDE) solver. The simulated electric-field intensity of the fundamental mode is shown in Fig. 1.

The wavelength is swept from 1500 nm to 1600 nm to investigate the wavelength dependence of the fundamental quasi-TE mode. The effective and group indices are obtained from the MODE frequency-sweep simulation, as shown in Fig. 2.

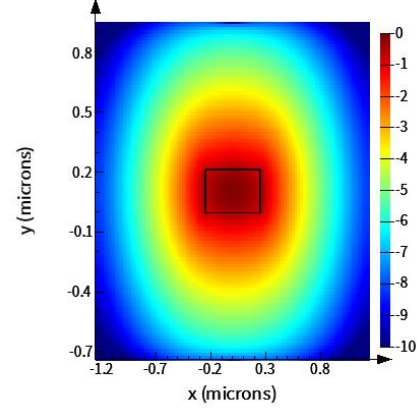


Fig. 1. Electric-field intensity (log scale) of the fundamental quasi-TE mode.

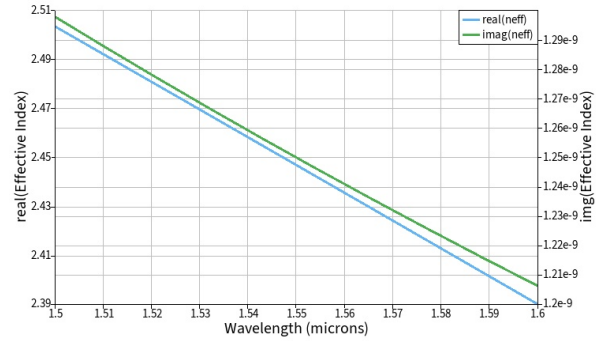


Fig. 2. Simulated effective and group index versus wavelength.

The simulated effective index is then fitted using MATLAB to obtain a second-order polynomial compact model,

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad (5)$$

where $\lambda_0 = 1550$ nm.

The simulated real part of n_{eff} is fitted using a second-order Taylor expansion around $\lambda_0 = 1.55$ μm :

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2. \quad (6)$$

The fitted coefficients are $n_1 = 2.4468$, $n_2 = -1.1334 \mu\text{m}^{-1}$, and $n_3 = -0.04393 \mu\text{m}^{-2}$. The resulting compact model is

$$n_{\text{eff}}(\lambda) = 2.4468 - 1.1334(\lambda - 1.55) - 0.04393(\lambda - 1.55)^2. \quad (7)$$

At λ_0 , the corresponding group index is

$$n_g = n_1 - n_2\lambda_0 = 4.2046, \quad (8)$$

B. MZI Modelling

An unbalanced Mach–Zehnder interferometer (MZI) is designed using the $500 \text{ nm} \times 220 \text{ nm}$ silicon strip waveguide described in the previous section. The two arms have a path-length difference $\Delta L = L_2 - L_1$. For an ideal 50/50 splitter and combiner, the transmission of the MZI can be expressed as

$$T_{\text{MZI}}(\lambda) = \frac{1}{2} \left[1 + \cos \left(\frac{2\pi n_{\text{eff}}(\lambda) \Delta L}{\lambda} \right) \right]. \quad (9)$$

The free spectral range (FSR) of the interferometer is given by

$$\text{FSR} \approx \frac{\lambda^2}{n_g \Delta L}. \quad (10)$$

Different path-length differences are considered to obtain suitable FSRs within the 1500–1600 nm wavelength range. For each ΔL , the expected FSR is calculated using eq. 10 where $\lambda_0 = 1550 \text{ nm}$ and n_g is obtained from the waveguide MODE simulation. The calculated FSR values for the selected ΔL values are listed in Table I.

TABLE I
CALCULATED FSR FOR DIFFERENT MZI PATH-LENGTH DIFFERENCES AT $\lambda_0 = 1550 \text{ nm}$.

$\Delta L \text{ (}\mu\text{m)}$	FSR (nm)
100	5.7140
120	4.7616
140	4.0814
160	3.5712
180	3.1744
200	2.8570

The MZI is implemented in Lumerical INTERCONNECT. The waveguide effective index and propagation loss are obtained from the MODE simulation. A three-port Y-branch S-parameter model from the course materials (S_YBranch.txt) is used for the splitter and combiner. A loss coefficient of 300 dB/m is used.

The MZI transmission spectrum is simulated using the Optical Network Analyzer over 1500–1600 nm. A representative spectrum for $\Delta L = 100 \mu\text{m}$ is shown in Fig. 3. The FSR is obtained from the wavelength spacing between adjacent transmission maxima.

The FSR is also extracted over the wavelength range and plotted as a function of wavelength in Fig. 4.

At $\lambda_0 = 1550 \text{ nm}$, the group index is extracted from the simulated FSR using Eq. 10. The results are summarized in Table II.

TABLE II
SIMULATED FSR AND EXTRACTED GROUP INDEX AT $\lambda_0 = 1550 \text{ nm}$.

$\Delta L \text{ (}\mu\text{m)}$	FSR (nm)	Extracted n_g
100	5.67906	4.2305
120	4.73378	4.2294
140	4.07355	4.2127
160	3.56849	4.2078
180	3.17780	4.2001
200	2.83478	4.2375

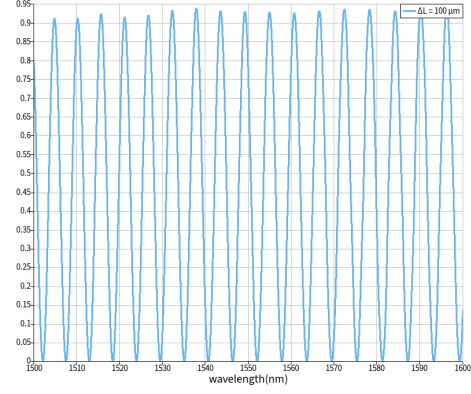


Fig. 3. Simulated MZI transmission spectrum for $\Delta L = 100 \mu\text{m}$.

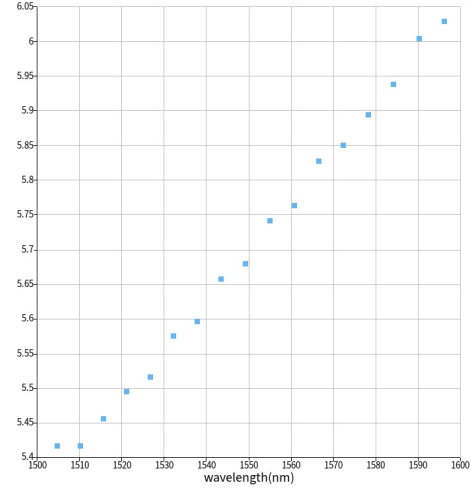


Fig. 4. Simulated FSR as a function of wavelength for $\Delta L = 100 \mu\text{m}$.

The extracted n_g values are approximately 4.22, which is consistent with the value obtained from the MODE simulation in Eq. 8.

IV. CONCLUSION

The MZI design was evaluated for several path-length differences to identify a suitable configuration for experimental group-index extraction. Based on the simulated FSR and the available measurement bandwidth, $\Delta L = 100 \mu\text{m}$ is selected because it provides the largest FSR and therefore the clearest spectral fringes within the measurement range. The extracted group index shows good agreement with the MODE simulation, validating the proposed FSR-based extraction approach. The selected MZI will be used for the final GDS layout and subsequent experimental characterization.